Training-Free Self-Calibrating Control Layer for Heterogeneous Multi-Object Trackers

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Abstract

Multi-object tracking systems combine a detector, which finds objects with confidence scores, with a tracker, which links them over time using fixed score thresholds. When the detector is replaced, retrained, or compressed, its scores change, but the tracker keeps its old thresholds. This paper proposes a training-free self-calibrating control layer (SCL) placed between an unchanged detector and tracker (the host). It splits the scores of recent frames with Otsu's histogram threshold and labels their distribution as clean or noisy. On clean scores it keeps the host's scores; on noisy scores it chooses which detections reach the host, rescores them, and sets its thresholds. It also continues tracks through low-score frames for single-stage trackers and needs no labels or future frames. One fixed configuration was tested on seven published trackers, four detection sources, and three datasets. Gains below are relative changes over the tracker alone. SCL raises the higher order tracking accuracy (HOTA) of two single-stage trackers by 0.9%: OC-SORT, used during development, and PD-SORT, which was not. With halved detector scores, two trackers failed; with SCL, both recovered at least 97% of their original HOTA. With a noisy RT-DETR-L detector, SCL raises the multiple object tracking accuracy (MOTA) of ByteTrack and BoT-SORT by 12.6% and 16.3% and reduces their identity switches by 43% and 47%. On four trackers that already fit their detections, HOTA changed by under 0.1%. Paired bootstrap analysis supports these results. SCL adds 2.95 milliseconds per frame on an ordinary processor.

Keywords: multi-object tracking, tracking by detection, detector confidence calibration, online adaptation, Otsu thresholding, real-time image processing

1 Introduction

Multi-object tracking (MOT) finds every object of interest in a video and gives each one an identity number (ID) that stays the same over time. It is used in surveillance, driving, and drone systems [1, 2], where results are needed with little delay. Most MOT systems follow the tracking-by-detection design [3]: in each frame, a detector returns candidate boxes with confidence scores, and a tracker links them into tracks, one per object [4–6]. The tracker uses a box only if its score passes a fixed threshold. A two-stage tracker, such as ByteTrack, also uses lower-score boxes in a second step to continue unmatched tracks [5];

a single-stage tracker, such as OC-SORT, ignores every box below one threshold [6].

In real systems the detector is often replaced, retrained, or compressed, while the tracker keeps the thresholds that its authors set for another detector. Detector scores are not calibrated across models and conditions [7], so a new detector usually gives scores on a different scale, and nothing in the system notices. The effect can be large: halving the scores of a published detector reduces two well-known trackers to 0 higher order tracking accuracy (HOTA) [8], because no detection passes their thresholds (Sect. 5.3). For example, ByteTrack in its official setting starts a track only from a box with a score of at least 0.7; a car that the original detector scored 0.9 now gets 0.45, so ByteTrack never starts a track, and its HOTA falls from 67.70 to 0. SCL finds from the scores themselves that the confident group now begins lower and lets it reach the tracker again, and ByteTrack reaches 66.17 HOTA again, with no labels and no change to its code. Retuning every tracker for every detector needs labels, which a working system usually does not have. Adaptive-threshold methods are each built into one tracker [9–11], score calibration needs labelled data [7], and adaptive inference methods learn their settings for one model [12, 13].

This paper presents a self-calibrating control layer (SCL) that sits between a detector and a tracker that are both kept fixed (frozen): their code, weights, and stored settings are not changed. We call the tracker that SCL serves the *host*; it gives SCL four of its published settings, the host contract (Sect. 3.7). SCL checks how many of the higher scores of recent frames form a clearly confident group, using only the scores, so it needs no labels. If most are confident, the host keeps its scores and, unless they would reject the confident group, its thresholds, and the layer only adds a step that continues tracks. If many are uncertain, SCL chooses which candidates reach the host, gives them new scores, and sets the host's thresholds for the frame.

With halved detector scores, SCL restored two trackers from 0 to about 66 HOTA; with other detectors, it raised the multiple object tracking accuracy (MOTA) of two-stage trackers by up to 24.7 points and cut their identity switches by 43% to 56%, while well-matched strong trackers stayed within 0.06 HOTA points of their own accuracy.

The main contributions are:

1. SCL, a training-free and causal layer that keeps the tracker's scores and thresholds when the scores are clean, lowering a threshold only if it would reject the confident group, and takes control only when they are noisy, with a host contract that lets one fixed configuration serve single-stage and two-stage trackers without changing them (Sect. 3).
2. A development path from always-on banding (AOB), an earlier design that always intervened, in which each step is compared with the step it was built on (Sect. 4).
3. An evaluation of one frozen configuration with paired bootstrap intervals, comparisons with simple adaptive thresholds and the published ACT method, an analysis of each part of the layer (Sect. 5), and a cost of about 3 ms per frame on an ordinary processor (central processing unit, CPU) (Sect. 6).

Section 2 covers the data, metrics, and related work; Sections 3 and 4 present SCL and its development; Section 5 gives the results; and Sections 6 and 7 discuss and conclude.

2 Background and Related Work

2.1 Datasets

Three public datasets are used. *MOT17* [14] contains 14 sequences of people filmed with static and moving cameras. As is common, the second half of each of the seven training sequences, called val-half (2,652 frames), is used for evaluation [5], with the published YOLOX-X detections [15] at the two published emission floors, the lowest scores emitted, of 0.01 and 0.1. To study score changes directly, these scores were also changed before the tracker by four transforms that keep their order: cubing (s^3), halving ($0.5\,s$), and logit temperatures $T = 2$ and $T = 0.5$, which divide the logit (log-odds, Sect. 3.1) of each score by T. *KITTI tracking* [16] has 21 training sequences (8,008 frames) recorded from a car and is scored on cars and pedestrians; here the trackers run with two other detectors, YOLOv8n [17] and RT-DETR-L [18], and keep thresholds that were set for other detectors. *UAVDT* [19] contains drone videos of vehicles; its 20 test sequences (16,592

frames) are used with ByteTrack, YOLOv8n, and RT-DETR-L.

MOT17 with four trackers, KITTI, and the score transforms were development data, looked at while SCL was designed. Two predeclared external trackers, PD-SORT and Hybrid-SORT, were chosen in advance and run once with the frozen configuration on the detections released by their authors on MOT17 val-half. The held-out UAVDT test split was also run once, after a written plan. Each external host was checked against its authors' number: EXACT within 0.2 HOTA points, CLOSE within 1.0.

2.2 Evaluation metrics

HOTA [8], the main metric, balances finding the objects and keeping their identities,

$$\mathrm{HOTA} = \frac{1}{19} \sum_{\alpha} \sqrt{\mathrm{DetA}_\alpha \, \mathrm{AssA}_\alpha}, \tag{1}$$

where DetA_α and AssA_α are the detection and association accuracy at the localization threshold $\alpha = 0.05, 0.10, \ldots, 0.95$. MOTA [20] and IDF1 [21] are

$$\mathrm{MOTA} = 1 - \frac{\mathrm{FN} + \mathrm{FP} + \mathrm{IDS}}{\mathrm{GT}}, \tag{2}$$

$$\mathrm{IDF1} = \frac{2\,\mathrm{IDTP}}{2\,\mathrm{IDTP} + \mathrm{IDFP} + \mathrm{IDFN}}, \tag{3}$$

where FN are missed objects (false negatives), FP false positives, IDS identity switches (an object's ID changes), and GT the number of ground-truth boxes; IDTP, IDFP, and IDFN count boxes after each ground-truth track is matched to at most one output track. All metrics are computed with TrackEval (commit 12c8791) and the official ground truth [22]; KITTI HOTA is averaged over cars and pedestrians, and on UAVDT all vehicles form one class and predictions in ignored regions are removed. HOTA, MOTA, and IDF1 are in percent, and differences in points.

All comparisons are paired: each tracker runs alone and with SCL on the same sequences and detections. Paired bootstrap resampling of whole sequences [23], 10,000 times with seed 42, gives a 95% interval (2.5–97.5 percentiles) for each difference, which is called significant when its interval does not include zero.

2.3 Related work

Tracking by detection. SORT matches Kalman-filter predictions to new detections by their overlap [4]; ByteTrack added a second step that uses low-score detections to continue tracks [5]; OC-SORT uses one score threshold [6]. Their thresholds were tuned for one detector, usually YOLOX [15]. Many recent trackers improve association with trained or domain-specific parts [24–29], which change or retrain the tracker; SCL keeps both the tracker and the detector unchanged. Seven published trackers serve as hosts in this study, each with the authors' code and published configuration: ByteTrack, BoT-SORT [30], OC-SORT, SparseTrack [31], BoostTrack [32], PD-SORT [33], and Hybrid-SORT [34]. ByteTrack is used with the thresholds that its authors used on MOT17 (official setting) and with the defaults of the Ultralytics library, which also provides BoT-SORT (library setting).

Adaptive thresholds. Adaptive ByteTrack sets ByteTrack's threshold from each frame [9], AdapTrack adapts its matching thresholds [10], and ACT sets the high and low thresholds from the first quartile and the number of the current scores [11]. ConfTrack weights the Kalman update and the matching cost by the score [35], UTrack uses the spread of the boxes merged by non-maximum suppression [36], and PlugTrack blends motion predictors with a trained network [37]. These methods are built into one tracker, use only the current frame, or need trained parts. SCL differs in four ways: it uses past frames; it decides for the whole video whether to act at all; it serves several unchanged trackers through their contracts; and it is tested as one frozen configuration under score changes and on external systems.

Score calibration and adaptive inference. Detector families such as YOLOX, YOLOv8, YOLOv9 [38], DETR [39], and RT-DETR give different score distributions, and calibration methods learn a mapping from labelled data [7]. Otsu's method picks, without labels, the threshold that best splits a histogram into two groups [40]; SCL applies it to the logits of recent scores. Dynamic networks [12] and Chameleon [13] learn or measure their settings for each model, and our earlier controller, AC-MOT (submitted), sets the detector operating point from image and track cues.

SCL instead reads the scores and acts on the tracker's input and thresholds.

3 Proposed SCL Framework

SCL is a small, rule-based layer that runs once per frame between a frozen detector and a frozen tracker, the host, and changes neither. Figure 1 shows its five steps for frame t, and this section follows them in order. The top row is the path of the detections: candidate choice (Step 3) keeps, drops, or rescores them, continuation rescue (Step 4) passes weak boxes that continue tracks, and the tracker matches them as usual. The bottom row sets this path: score statistics (Step 1) and the mode check (Step 2) tell whether the scores are clean or noisy, the host contract gives the tracker's published settings, and the matching tolerance (Step 5) follows the camera motion.

3.1 Inputs and outputs

In each frame t, the detector gives candidates $D_t = \{(\mathbf{x}_i, s_i, c_i)\}$, each with a box $\mathbf{x}_i$, a score $s_i \in (0, 1)$, and a class c_i. The layer works on the logit

$$z_i = \log \frac{s_i}{1 - s_i}, \tag{4}$$

the log-odds transformation of the confidence score. It maps a score between 0 and 1 to an unbounded scale and keeps the order of the scores: a score of 0.5 gives $z_i = 0$, 0.9 about 2.2, and 0.1 about -2.2. The logistic function $\sigma(z) = 1/(1 + e^{-z})$ turns a logit back into a score. The layer uses logits for two reasons. Detector scores crowd near 0 and 1, where small differences in score are large differences in confidence; the logit spreads these regions out, so the splits of Step 1 separate the groups more clearly. And a common change of a detector's scores, a change of temperature, divides all logits by the same number; the splits follow this stretch, and each group keeps the same candidates. The host gives its contract $h = (a, b, \ell, m_0)$: the score a needed to match a track, the score b needed to start one, the lowest score ℓ it uses, and its matching tolerance m_0, one minus the smallest intersection over union (IoU) of two boxes that counts as a match. A single-stage tracker has $\ell = \min(a, b)$, a two-stage tracker $\ell < \min(a, b)$. In every frame the layer returns

$$(K_t, \tilde{s}, a_t, b_t, m_t) = \mathcal{L}\left(Z_{<t}, D_t, \mathcal{T}_{t-1}, g_t; h\right): \tag{5}$$

the candidates K_t for the tracker, their scores, and its thresholds and tolerance (the arrows into the tracker in Fig. 1), from the logits $Z_{<t}$ of earlier frames, the tracks $\mathcal{T}_{t-1}$, and the camera motion g_t between the frames I_{t-1} and I_t; no future frame is used.

3.2 Step 1: score statistics

Input: the logits Z_t^W of the candidates of the last $W = 10$ frames. *Output:* the splits t_1, t_2 and the confident share ρ_t.

The logits are split twice with Otsu's method [40], which picks, from the data alone and without labels or tuning, the threshold that best separates a set of values into two groups; applied twice, it gives the three groups that one threshold cannot. The first split, t_1, separates background (low logits) from foreground; the second, t_2, found on the foreground only, divides it into uncertain and confident candidates. The share of confident candidates in the foreground,

$$\rho_t = \frac{\left|\{z \in Z_t^W : z \geq t_2\}\right|}{\left|\{z \in Z_t^W : z \geq t_1\}\right|}, \tag{6}$$

describes the shape of the score distribution. For example, for the scores 0.02, 0.03, 0.05, 0.4, 0.5, 0.9, and 0.95 (logits -3.9, -3.5, -2.9, -0.4, 0, 2.2, and 2.9), the first split separates the three lowest scores as background, and the second divides the other four into 0.4 and 0.5 (uncertain) and 0.9 and 0.95 (confident), so $\rho_t = 2/4 = 0.5$. Figure 2 shows the groups for a clean and a noisy stream. When a detector emits very few low scores, the first split falls inside the objects; since clutter usually gives more candidates than the objects do, a window with fewer candidates below t_1 than above it counts as clean, $\rho_t = 1$ (background check).

3.3 Step 2: mode check

Input: the values ρ of all frames so far. *Output:* the mode of frame t.

The mode describes the distribution of the scores, not whether a single detection is correct. The stream is clean at frame t if the median of all ρ values so far is at least one half, a majority rule, and noisy otherwise; before the first valid split it is cold. Because the median covers the whole stream, a short drop in confidence does not change the

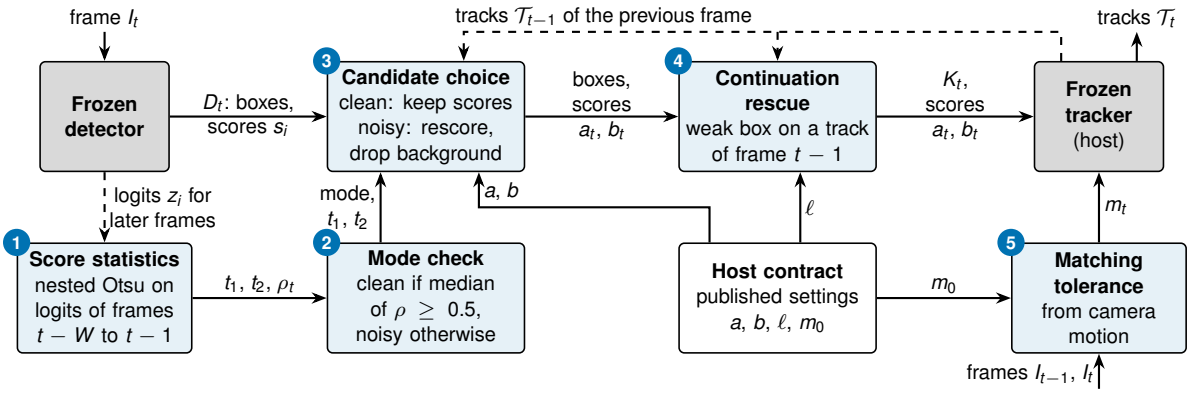


Fig. 1 SCL between a frozen detector and a frozen tracker (the host), for frame t. Each arrow is labelled with the data it carries; dashed arrows carry data kept from earlier frames. Steps 1–5 are described in Sects. 3.2–3.6, and the host contract in Sect. 3.7

mode (Sect. 4.2). The splits themselves use only the last W frames, so the groups, and in the clean mode the thresholds of Step 3, follow a change of the score scale within W frames even while the mode stays the same.

3.4 Step 3: candidate choice

Input: D_t, the mode, t_1, t_2, the thresholds a, b, and the tracks $\mathcal{T}_{t-1}$. *Output:* the kept boxes, their scores, and the thresholds a_t, b_t.

Cold and clean modes. The tracker keeps its own thresholds and scores, and only cross-class duplicates are removed. In the clean mode, a threshold so high that it would reject the confident group is lowered to the score $\sigma(t_2)$ at which that group begins:

$$a_t = \min(a, \sigma(t_2)), \qquad b_t = \min(b, \sigma(t_2)). \quad (7)$$

The clean mode is thus a minimal intervention rather than a plain pass-through: the scores are kept, and the output equals that of the host alone whenever the confident group lies above the host's thresholds and no rescue applies, as for BoostTrack, ByteTrack at floor 0.1, and Hybrid-SORT on MOT17 (Sect. 5). When the scores are halved, for example, no score is above 0.5, and ByteTrack in its official setting ($a = 0.6$, $b = 0.7$) never starts a track; the splits move with the halved scores, so Eq. (7) lets the confident candidates reach it again.

Noisy mode. Background candidates ($z < t_1$) are dropped. Each remaining score is replaced by its mid-rank $u_i \in (0, 1)$ among the scores of a remap memory, which stores the scores kept in the first frame and in every 10th frame, the last 20 of them (tied scores share their average rank, and before the first sample the current frame is used),

$$\tilde{s}_i = \begin{cases} 0.5 + 0.5\,u_i, & z_i \geq t_2, \\ 0.1 + 0.4\,u_i, & t_1 \leq z_i < t_2, \end{cases} \quad (8)$$

and both thresholds are set to $a_t = b_t = 0.5$. Confident candidates ($z \geq t_2$) then get scores above 0.5 and may start and continue tracks, and uncertain ones ($t_1 \leq z < t_2$) get scores below 0.5 and may only continue tracks through a tracker's low-score step. A weaker candidate that overlaps a stronger one with IoU above one half is removed, unless a different track of the previous frame claims it, which keeps people who stand close together in a crowd.

3.5 Step 4: continuation rescue

Input: the output of Step 3, the tracks $\mathcal{T}_{t-1}$, and ℓ. *Output:* K_t, their scores, and a_t, b_t for the tracker.

A track can be lost when its object gets low scores for a few frames, above all in single-stage trackers, which have no low-score step. The rescue passes such a candidate, with its score raised just above the lowest threshold of the frame (ℓ for a two-stage tracker, the lower of a_t and b_t for a single-stage one), when it is in the foreground (or, for a single-stage tracker, when there is no background group), overlaps a track of the previous frame with IoU of at least one half, no other usable candidate covers that track, and the tracker

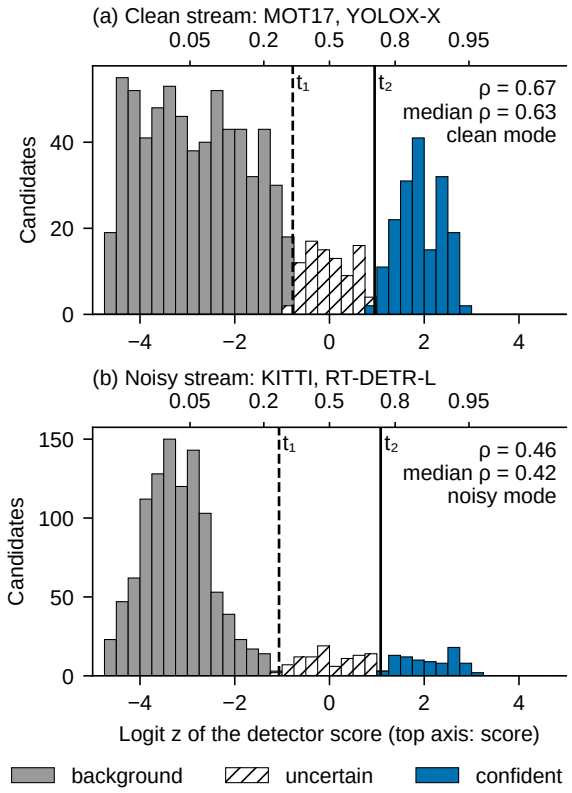


Fig. 2 What the layer sees: the logits of the candidates of the last 10 frames, just before its decision for frame 101, with the Otsu splits t_1 (dashed line) and t_2 (solid line). Plain light bars: background; hatched bars: uncertain; plain dark bars: confident. The top axis gives the matching detector score. (a) MOT17-02 with YOLOX-X: most foreground candidates are confident, and the stream is clean. (b) KITTI 0001 with RT-DETR-L: many are uncertain, and the stream is noisy. Values of the frozen layer in the runs of Sect. 5

could not see it otherwise. It thus gives single-stage trackers the continuation step they lack. The rescue only makes the box visible; the tracker itself still decides, with its own matching, whether and to which track it is assigned.

3.6 Step 5: matching tolerance

Input: the frames I_{t-1}, I_t and the tolerance m_0. *Output:* the tolerance m_t of frame t.

In noisy frames, the matching tolerance grows with camera motion, because fast motion moves boxes far from their tracks; otherwise $m_t = m_0$. The motion g_t is the shift between consecutive grayscale frames at quarter resolution, found by phase correlation [41], and r_t is its ratio to its median over the last 100 frames ($r_t = 1$ during a warm-up of 5 frames or if this median is zero):

$$m_t = \min\left(0.95,\ 1 - \frac{1 - m_0}{\max(1, r_t)}\right). \qquad (9)$$

With $m_0 = 0.8$ and twice the usual motion, a match needs an IoU of 0.1 instead of 0.2.

Table 1 Host contracts: the published settings of each tracker, as used by the layer: matching (association) threshold a, birth threshold b, low threshold ℓ, and match tolerance m_0. For every tracker, the layer's decision is passed on as thresholds, candidates, and scores.

Tracker (setting)	Type	a	b	ℓ	m_0
ByteTrack (official MOT17)	two-stage	0.6	0.7	0.1	0.8
ByteTrack (library default)	two-stage	0.25	0.25	0.1	0.8
BoT-SORT (library default)	two-stage	0.25	0.25	0.1	0.8
SparseTrack	two-stage	0.6	0.7	0.1	0.85
BoostTrack	two-stage	0.6	0.6	0.1	0.7
Hybrid-SORT	two-stage	0.6	0.6	0.1	0.75
OC-SORT	single-stage	0.6	0.6	0.6	0.7
PD-SORT	single-stage	0.6	0.6	0.6	0.7

3.7 Host contract and adapters

Table 1 lists the contracts; each value is taken from the tracker's published configuration, and none is fitted. For each tracker family, a small adapter links the layer to the tracker block of Fig. 1: it writes the layer's candidates, scores, thresholds, and tolerance into the tracker's existing parameters before each update and then calls the tracker's own update. The matching steps, motion model, track management, and any post-processing of the authors run unchanged, so the host stays frozen: only the values of its thresholds are set for each frame, and the rules are the same for every tracker.

3.8 Parameter values

Table 2 lists all values of SCL. They were fixed before the external evaluation, are the same for every tracker, detector, and dataset, and none was chosen by searching over a range of values. They come from standard values (the IoU of 0.5 of MOT evaluation and the common thresholds 0.5 and 0.1), from definitions, and from design choices kept fixed in all development stages. Algorithm 1 gives Steps 1–5 for one frame.

Table 2 Values of SCL and their source.

Value	Setting	Source
Window W	10 frames	design choice
Mode rule	clean if median $\rho \geq 0.5$	majority rule
Background check	$\rho = 1$ if fewer below t_1 than above	rule: no background group
Noisy score ranges	$[0.5, 1)$, $[0.1, 0.5)$	common thresholds
Noisy a_t, b_t	0.5	common threshold
Remap memory	20 frames, one every 10	design choice
Duplicate, rescue IoU	0.5	MOT evaluation IoU
Rescued score	lowest threshold $+0.001$	just above the threshold
Motion cue	phase correlation, quarter resolution	standard method [41]
Motion history	100 frames, warm-up 5	design choice
Tolerance cap	0.95	minimum overlap 0.05

4 Development of the Final SCL

SCL was developed in three stages on the development data of Sect. 2.1: a starting point, design steps that each removed one cause found before them (Table 3), and a freeze before any external test.

4.1 Starting point: always-on banding

The earlier design AOB computed the same nested Otsu groups but treated every stream as noisy: in every frame it applied the groups, removed every weaker candidate that overlapped a stronger one with IoU above one half, and rescored all candidates by rank. Table 4 compares AOB and SCL with strong published trackers on MOT17. AOB lowered the HOTA of SparseTrack by 4.15 points and of BoostTrack by 5.89 points online and 5.37 points with its gap-filling post-processing (GBI), for two reasons: removing overlapping boxes deleted people hidden behind others in crowds, and its thresholds were stricter than the trackers' own. With the same trackers, SCL changed HOTA by +0.06 for SparseTrack and left both BoostTrack versions identical to the host. A layer that acts in every frame therefore cannot be placed safely in front of an unknown tracker, and the next steps let the layer decide from the scores whether to act at all.

Algorithm 1 SCL for frame t; the comments name the steps of Fig. 1. The window holds the logits of frames $t-W$ to $t-1$, so the decision for frame t uses no future frame.

Input: candidates D_t, tracks $\mathcal{T}_{t-1}$, motion cue g_t, contract (a, b, ℓ, m_0), constants of Table 2
Output: candidates K_t, scores $\tilde{s}$, thresholds a_t, b_t, tolerance m_t

$z_i \leftarrow$ logit of s_i for all i ▷ Eq. (4)
if no valid split exists **then** ▷ e.g., first frame
  mode $\leftarrow$ cold; $(a_t, b_t) \leftarrow (a, b)$
else
  $(t_1, t_2) \leftarrow$ nested Otsu on the window ▷ Step 1
  $\rho_t \leftarrow$ Eq. (6) with the background check
  mode $\leftarrow$ clean if median$(\rho) \geq 0.5$, else noisy ▷ Step 2
end if
if mode is cold or clean **then** ▷ Step 3
  remove cross-class duplicates; keep raw scores
  if clean **then** set (a_t, b_t) by Eq. (7)
  end if
else ▷ noisy mode
  remove duplicates with track context
  drop $z < t_1$; remap scores by Eq. (8)
  $a_t \leftarrow 0.5$; $b_t \leftarrow 0.5$
end if
rescue continuations (not cold) ▷ Step 4
$m_t \leftarrow$ Eq. (9) if noisy, else m_0 ▷ Step 5
pass K_t, $\tilde{s}$, a_t, b_t, m_t to the tracker
add the logits left after duplicate removal to the window
add g_t to the motion history
store the tracks of frame t after the tracker update

4.2 Design steps

Each step was compared with the step before it (Table 3). The *mode gate* added the mode check, the clean mode that keeps the tracker's scores and thresholds, and the crowd-safe duplicate rule; it restored SparseTrack from 64.72 to 68.93 HOTA and BoostTrack from 62.61 to 68.37. The *whole-stream mode* removed mode changes in the middle of streams and raised ByteTrack from 67.561 to 67.702 and OC-SORT from 65.697 to 66.071. The *continuation rescue* added 0.2 to 0.4 points for OC-SORT. The *background check* completed SCL: ByteTrack and BoostTrack returned to output

Table 3 Design steps from AOB to SCL on development data (HOTA, %), and versions that were tested and rejected. The floor of ByteTrack and OC-SORT is given in parentheses.

Step	Change	Effect
AOB emulated	always-noisy bands, overlap deduplication, rank remap	ByteTrack (0.1) 56.41 vs host 67.70; OC-SORT (0.1) 45.72 vs 66.44
+ mode gate	mode statistic, host-anchored clean mode, crowd-safe duplicates	SparseTrack 64.72 to 68.93 (host 68.88); BoostTrack 62.61 to 68.37 (host 68.49)
+ whole-stream mode	mode from all past frames	ByteTrack (0.1) 67.561 to 67.702; OC-SORT (0.1) 65.697 to 66.071
+ continuation rescue	foreground track-consistent rescue	OC-SORT (0.01) 66.405 to 66.619; OC-SORT (0.1) 66.071 to 66.469
+ background check	no background mode gives	ByteTrack/BoostTrack (0.1) identical to host;
= SCL	no evidence against the host	OC-SORT 67.041 (0.01), 66.907 (0.1)
Rejected version		*Evidence*
no candidates at frame 1		−0.3 to −0.5 HOTA on every MOT17 host
rescue below a two-stage host's low stage		FP +700 to +1000, −0.4 to −0.5 HOTA
rank-remapped scores in clean frames		ByteTrack −0.2 / −1.6 HOTA
host-clipped noisy primary band		KITTI YOLOv8n ByteTrack +1.4 HOTA but uncontrolled false tracks on a noisy aerial stream (26.9 vs 14.9 boxes/frame)

Table 4 Baseline comparison of the development stage on MOT17 val-half (published YOLOX-X detections). Host: the tracker alone, without any layer (our run of the published tracker). Δ: tracker with the earlier design AOB or with SCL minus host, with 95% paired-bootstrap interval (10,000 resamples, seed 42). Bold intervals exclude zero. IDS: identity switches, a count (lower is better).

Metric	Host	Δ AOB	Δ SCL
SparseTrack [31]			
HOTA	68.88	**−4.15** [**−5.54**, **−1.66**]	+0.06 [−0.01, +0.25]
MOTA	77.85	**−6.14** [**−8.10**, **−2.74**]	+0.08 [−0.03, +0.30]
IDF1	81.97	**−4.49** [**−6.19**, **−1.96**]	+0.16 [−0.02, +0.70]
IDS	124	−16 [−47, +12]	+7 [0, +19]
BoostTrack, online [32]			
HOTA	68.49	**−5.89** [**−7.76**, **−2.87**]	0 (identical)
MOTA	75.50	**−8.87** [**−12.37**, **−7.14**]	0 (identical)
IDF1	81.41	**−6.22** [**−9.33**, **−2.85**]	0 (identical)
IDS	113	**−38** [**−81**, **−1**]	0 (identical)
BoostTrack, with GBI [32]			
HOTA	71.72	**−5.37** [**−7.30**, **−2.75**]	0 (identical)
MOTA	81.03	**−8.73** [**−12.59**, **−6.50**]	0 (identical)
IDF1	84.16	**−5.65** [**−8.96**, **−2.77**]	0 (identical)
IDS	97	−27 [−73, +11]	0 (identical)

identical to the host, and OC-SORT rose to 67.04 and 66.91 at the two floors. Four other variants were tested and not kept.

4.3 Freezing and testing

The final configuration was frozen before any external test as a version-control tag, protected by a lock file with a cryptographic hash of the layer, its configuration, and the evaluation scripts. Sixty-one automated tests check, among others, that the layer uses no future data and passes clean streams unchanged.

5 Results

A tracker that fits its detector should keep its accuracy, and one that does not, because it lacks a continuation step or because the detector or its scores changed, should gain. For each host, the baseline is the host alone (Host in the tables), run with its published configuration and the same detections; the second run adds the frozen SCL and changes nothing else. Figure 3 summarizes the HOTA difference for every tested cell.

5.1 MOT17 evaluation across trackers

Table 5 shows the development trackers on MOT17 val-half. The two-stage trackers with well-matched detections stayed at the accuracy of their hosts: BoostTrack, online and with GBI, and ByteTrack at floor 0.1 gave exactly the same output, and ByteTrack at floor 0.01 and SparseTrack

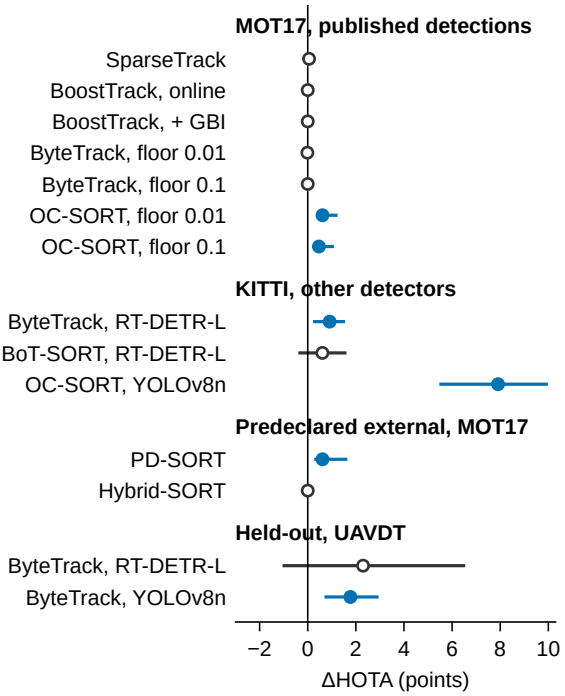


Fig. 3 Difference in HOTA, host with SCL minus host alone, in points, for every tested cell, with its 95% paired-bootstrap interval. Filled circles: interval above zero; open circles: interval contains zero, or identical output

stayed within 0.06 HOTA points, the self-limiting clean mode. The single-stage OC-SORT improved at both floors (HOTA +0.61 and +0.46, MOTA +1.27 and +1.26), with fewer identity switches and higher HOTA on 7 of 7 sequences, through the continuation rescue.

5.2 KITTI evaluation with other detectors

Table 6 shows what happens when the detector is replaced. With the noisy RT-DETR-L, ByteTrack gained 5.98 MOTA, 3.87 IDF1, and 0.91 HOTA points, and BoT-SORT 7.86 MOTA and 3.10 IDF1 points, with 297 and 226 fewer identity switches. With YOLOv8n, OC-SORT gained 7.91 HOTA points ([+5.47, +9.98]), 9.40 MOTA points, and 9.60 IDF1 points; it followed many more objects, as its missed objects (FN) fell by 5,310, and its identity switches rose with them by 79. Figure 4 follows this cell over three frames: each object keeps its ID, and at frame 102 OC-SORT alone tracked 6 of the 10 labelled objects and with SCL 9.

Table 5 Baseline comparison of the development trackers on MOT17 val-half (published YOLOX-X detections). Paper: number reported by the tracker's authors. Host: the tracker alone, without SCL (our run of the published tracker). Δ: host with SCL minus host, with 95% paired-bootstrap interval (10,000 resamples, seed 42); bold intervals exclude zero. The floor is the lowest detector score emitted. IDS: identity switches, a count (lower is better); –: not reported by the authors.

Metric	Paper	Host	Δ [95% CI]
SparseTrack [31]			
HOTA	69.2	68.88	+0.06 [−0.01, +0.25]
MOTA	76.8	77.85	+0.08 [−0.03, +0.30]
IDF1	81.4	81.97	+0.16 [−0.02, +0.70]
IDS	–	124	+7 [0, +19]
BoostTrack, online [32]			
HOTA	68.37	68.49	0 (identical)
MOTA	75.56	75.50	0 (identical)
IDF1	81.35	81.41	0 (identical)
IDS	118	113	0 (identical)
BoostTrack, + GBI [32]			
HOTA	71.33	71.72	0 (identical)
MOTA	80.55	81.03	0 (identical)
IDF1	83.84	84.16	0 (identical)
IDS	106	97	0 (identical)
ByteTrack, floor 0.01 [5]			
HOTA	–	67.70	−0.014 [−0.058, +0.009]
MOTA	76.6	77.60	+0.06 [−0.15, +0.34]
IDF1	79.3	79.47	−0.031 [−0.135, +0.035]
IDS	–	214	+4 [−2, +13]
ByteTrack, floor 0.1 [5]			
HOTA	–	67.70	0 (identical)
MOTA	76.6	77.60	0 (identical)
IDF1	79.3	79.47	0 (identical)
IDS	–	214	0 (identical)
OC-SORT, floor 0.01 [6]			
HOTA	66.5	66.43	**+0.61 [+0.36, +1.24]**
MOTA	74.9	74.67	**+1.27 [+0.17, +3.01]**
IDF1	77.7	78.05	+0.66 [−0.25, +1.60]
IDS	–	211	**−8 [−15, −2]**
OC-SORT, floor 0.1 [6]			
HOTA	66.5	66.44	**+0.46 [+0.27, +1.09]**
MOTA	74.9	74.67	**+1.26 [+0.37, +3.34]**
IDF1	77.7	78.05	**+0.287 [+0.003, +0.838]**
IDS	–	213	−14 [−38, +6]

5.3 Evaluation under shifted detector scores

Table 7 shows the trackers under the four score transforms. Halving the scores reduced ByteTrack (official setting) and OC-SORT to 0 HOTA; with the layer they reached 66.17 and 66.72. Cubing reduced them to 60.92 and 58.34, and the layer restored them to 65.81 and 66.60; BoostTrack rose from 61.33 to 64.84. With $T = 2$, the layer raised ByteTrack and OC-SORT by 1.11 and 1.24 points. The library setting of ByteTrack, whose low thresholds already accept the changed scores,

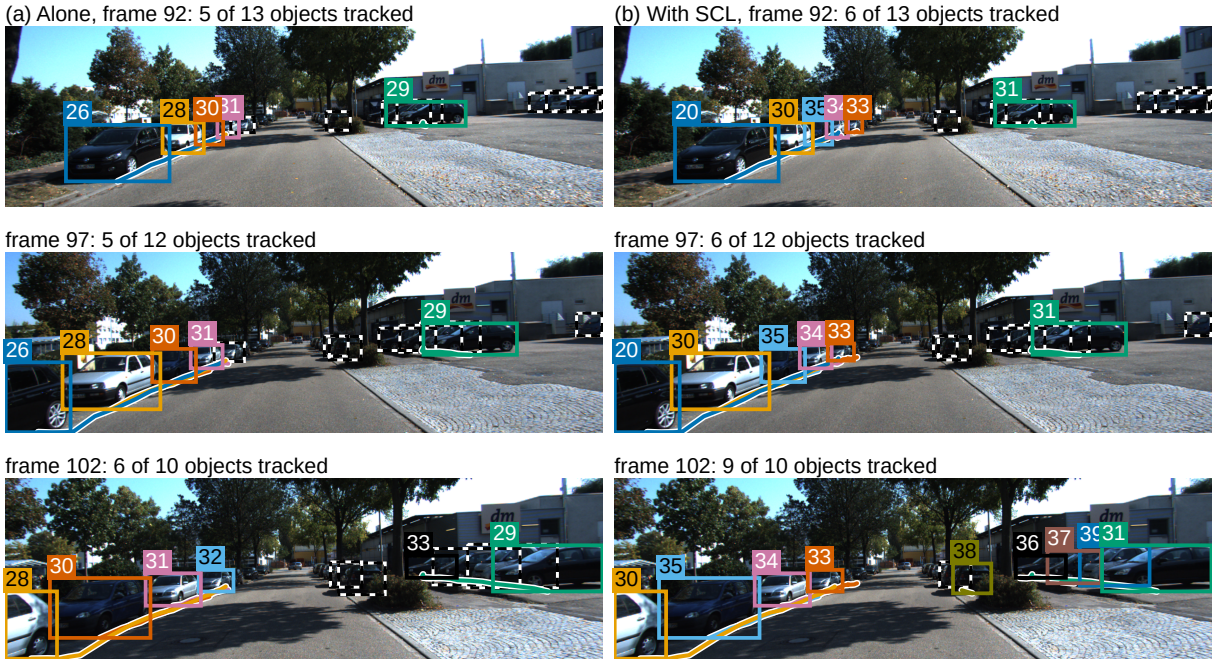


Fig. 4 Tracking on KITTI sequence 0001 [16] with OC-SORT and YOLOv8n, at frames 92, 97, and 102 (rows): (a) the tracker alone, (b) with SCL. Each output track keeps one ID and one colour over the frames; the line of the same colour is its path over the previous 20 frames (bottom centre of its box). Dashed boxes: labelled objects that no track covers. IDs are numbered separately in each run. The sequence has the median HOTA gain of the 21 sequences, and frame 102 is its first frame with the largest gain in tracked objects

stayed within 0.14 points. Over the 14 conditions, the layer raised HOTA significantly in 7 and lowered it significantly in none.

5.4 Comparison with adaptive-threshold methods

Two simple baselines set the tracker's matching and birth thresholds in every frame to the first or second Otsu split of the logits of the last 10 frames, t_1 or t_2, and pass the scores unchanged. On KITTI (Table 8), SCL reached the highest HOTA in all three cells, 2.2 to 8.6 points above the baselines, with intervals against the t_2 baseline above zero in every cell, so adaptive thresholds alone do not explain the gains. With OC-SORT, SCL also had fewer identity switches than both baselines (170 against 1,545 and 250 with YOLOv8n; 308 against 1,975 and 492 under halved scores).

ACT [11] was run with its published parameters, without retuning, on the same detections (Table 9). SCL reached higher HOTA than ACT in every row, significantly under halved scores (+2.01 and +10.06 points) and with YOLOv8n on UAVDT (+1.82), and also in the MOT17 setting for which ACT was designed. With RT-DETR-L on UAVDT, SCL reached a MOTA of 27.99 against 5.34 (+22.65, [+9.03, +47.21]), because the fixed score levels of ACT removed few of the host's false positives (178,219 against 78,975). SCL also had fewer identity switches than ACT on MOT17 and UAVDT (up to 418 fewer).

Removing one part at a time on KITTI shows where the gains come from: without the rescue, HOTA fell by up to 3.35 points; without the mode decision, by up to 2.44; and without the new scores of the noisy mode, by up to 1.67.

5.5 Frozen evaluation on external trackers

Table 10 gives the two predeclared external systems. PD-SORT [33] was reproduced exactly. With SCL, its HOTA rose by 0.61 points ([+0.27, +1.65]), its MOTA by

Table 6 Baseline comparison under a detector change on KITTI tracking training (official HOTA, car and pedestrian averaged). Host: the tracker alone, without SCL, with the same detections. Δ: host with SCL minus host, with 95% paired-bootstrap interval; bold intervals exclude zero. IDS: identity switches, a count (lower is better). BoT-SORT: KITTI sequence 0020 is left out of both runs, because the BoT-SORT code stopped with a numerical error in its Kalman update on it; the other 20 sequences are used.

Metric	Host	Δ [95% CI]
ByteTrack [5], *RT-DETR-L*		
HOTA	50.73	**+0.91 [+0.22, +1.55]**
MOTA	47.27	**+5.98 [+1.16, +9.94]**
IDF1	64.98	**+3.87 [+1.67, +5.06]**
IDS	686	**−297 [−530, −122]**
BoT-SORT [30], *RT-DETR-L* (20 sequences)		
HOTA	53.91	+0.61 [−0.39, +1.61]
MOTA	48.25	**+7.86 [+2.21, +13.44]**
IDF1	67.74	**+3.10 [+1.00, +4.57]**
IDS	476	**−226 [−369, −120]**
OC-SORT [6], *YOLOv8n*		
HOTA	36.27	**+7.91 [+5.47, +9.98]**
MOTA	33.99	**+9.40 [+1.10, +12.30]**
IDF1	50.46	**+9.60 [+5.61, +12.16]**
IDS	91	**+79 [+43, +125]**

Table 7 Baseline comparison under shifted detector scores on MOT17 val-half. The detector scores are transformed before the tracker, whose published thresholds stay fixed; T is a logit temperature, $\sigma(z/T)$. Host: the tracker alone, without SCL. Δ: with SCL minus host, with 95% paired-bootstrap interval (10,000 resamples, seed 42); bold intervals exclude zero. Emission floor 0.01 (BoostTrack 0.1).

	HOTA (%)		
Transform	Host	+ SCL	Δ [95% CI]
ByteTrack, official setting [5]			
s^3	60.92	65.81	**+4.89 [+2.60, +13.69]**
$0.5\,s$	0.00	66.17	**+66.17 [+49.32, +75.40]**
$T = 2$	65.84	66.95	**+1.11 [+0.35, +3.87]**
$T = 0.5$	67.30	67.29	−0.009 [−0.051, +0.005]
ByteTrack, library setting [5]			
s^3	66.31	66.31	−0.006 [−0.384, +0.383]
$0.5\,s$	66.61	66.48	−0.14 [−0.66, +0.14]
$T = 2$	63.97	63.97	0 (identical)
$T = 0.5$	66.53	66.52	−0.006 [−0.029, +0.001]
OC-SORT [6]			
s^3	58.34	66.60	**+8.26 [+5.86, +16.81]**
$0.5\,s$	0.00	66.72	**+66.72 [+50.32, +75.56]**
$T = 2$	65.99	67.22	**+1.24 [+0.52, +3.42]**
$T = 0.5$	66.61	67.06	+0.45 [−0.01, +0.97]
BoostTrack, online [32]			
s^3	61.33	64.84	**+3.51 [+0.85, +13.28]**
$T = 2$	67.59	67.59	0 (identical)

1.13 ([+0.27, +3.03]), and its IDF1 by 0.82 ([+0.43, +1.94]), with higher HOTA on 7 of 7 sequences; like OC-SORT, it is a single-stage tracker, but it was not seen during design. Hybrid-SORT [34], a two-stage tracker, gave identical output, as predicted before the run.

Table 8 HOTA (%) of SCL and of two simple adaptive-threshold baselines on KITTI tracking training (official HOTA, car and pedestrian averaged). Host: the tracker alone, without any layer. t_1, t_2: the adaptive baselines, in which the tracker's matching and birth thresholds are set in every frame to the first or second Otsu split of the logits of the last 10 frames, and the scores are passed unchanged. Halved: halved detector scores. Δ: SCL minus the t_2 baseline, with 95% paired-bootstrap interval below the value; bold intervals exclude zero.

Cell	Host	t_1	t_2	SCL	Δ
OC-SORT, YOLOv8n	36.27	39.97	41.09	44.17	**+3.09 [+2.00, +4.03]**
ByteTrack, RT-DETR-L, halved	49.50	49.58	49.50	51.78	**+2.29 [+0.28, +4.96]**
OC-SORT, YOLOv8n, halved	0.00	37.68	43.56	46.30	**+2.74 [+1.32, +3.34]**

5.6 Frozen UAVDT transfer

On the held-out UAVDT test split (Table 11), SCL raised the HOTA of the YOLOv8n stream by 1.77 points ([+0.70, +2.94]) and its IDF1 by 3.97 points, and cut identity switches by 45%. With RT-DETR-L, whose host produced many false positives (MOTA 3.29), SCL raised MOTA by 24.70 points ([+10.06, +51.08]) and IDF1 by 5.74 points, and cut false positives by 58% and identity switches by 56%. No HOTA decrease was detected with either detector.

6 Discussion

The score scale is part of the MOT system. Halving the scores reduced ByteTrack and OC-SORT from 67.70 and 66.43 HOTA to 0, and with RT-DETR-L on UAVDT ByteTrack reached a MOTA of only 3.29. The fixed thresholds of a tracker hold only for the detector they were tuned for, so the score scale deserves the same attention as the association method.

Self-limiting by design. In the clean mode the layer keeps the tracker's thresholds and scores, so a well-matched tracker receives its normal input; AOB, which always intervened, lowered the same

Table 9 HOTA of ACT [11] and of SCL on the same detections, and the differences SCL minus ACT in HOTA and IDF1, with 95% paired-bootstrap interval below each value; bold intervals exclude zero. ACT: published parameters, no tuning. KITTI: official HOTA, car and pedestrian averaged; MOT17: val-half, ACT's own setting; UAVDT: test split, one vehicle class, ignored regions removed (Sect. 2.2).

Data	Detector	Tracker	ACT	SCL	ΔHOTA	ΔIDF1
KITTI, halved scores	RT-DETR-L	ByteTrack	49.77	51.78	**+2.01** [**+0.01**, **+4.76**]	+1.70 [−1.13, +6.80]
KITTI, halved scores	YOLOv8n	OC-SORT	36.24	46.30	**+10.06** [**+6.88**, **+12.01**]	**+12.03** [**+6.85**, **+14.69**]
MOT17, floor 0.01	YOLOX-X	ByteTrack (official)	67.46	67.68	+0.22 [−0.43, +0.88]	+0.57 [−0.68, +2.12]
MOT17, floor 0.1	YOLOX-X	ByteTrack (official)	67.46	67.70	+0.23 [−0.40, +0.90]	+0.60 [−0.55, +2.10]
UAVDT test	YOLOv8n	ByteTrack	29.01	30.83	**+1.82** [**+0.65**, **+3.00**]	**+4.00** [**+1.85**, **+6.22**]
UAVDT test	RT-DETR-L	ByteTrack	40.14	42.37	+2.23 [−0.97, +6.32]	+5.35 [−0.03, +12.47]

Table 10 Baseline comparison on the predeclared external systems (MOT17 val-half, authors' released detections), each evaluated once with the frozen SCL. Paper: number reported by the authors (for PD-SORT, their released output). Host: the tracker alone, without SCL (our run of the published tracker). Reproduction class against the authors' number (EXACT: $|\Delta\text{HOTA}| \leq 0.2$; CLOSE: ≤ 1.0 with a documented, untuned cause). Δ: host with SCL minus host, with 95% paired-bootstrap interval; bold intervals exclude zero. IDS: identity switches, a count (lower is better); –: not reported.

Metric	Paper	Host	Δ [95% CI]
PD-SORT [33], *EXACT*			
HOTA	68.01	68.01	**+0.61** [**+0.27**, **+1.65**]
MOTA	75.19	75.19	**+1.13** [**+0.27**, **+3.03**]
IDF1	81.03	81.03	**+0.82** [**+0.43**, **+1.94**]
IDS	–	129	+2 [−9, +13]
Hybrid-SORT [34], *CLOSE*			
HOTA	67.1	66.70	0 (identical)
MOTA	75.8	75.52	0 (identical)
IDF1	78.0	77.56	0 (identical)
IDS	–	259	0 (identical)

Table 11 Baseline comparison on the held-out UAVDT test split (20 aerial sequences, not used during design), evaluated once with the frozen SCL. Host: ByteTrack alone, without SCL, with its published setting. Δ: host with SCL minus host, with 95% paired-bootstrap interval; bold intervals exclude zero. IDS: identity switches, a count (lower is better).

Metric	Host	Δ [95% CI]
ByteTrack, RT-DETR-L [5, 18]		
HOTA	40.07	+2.30 [−1.05, +6.55]
MOTA	3.29	**+24.70** [**+10.06**, **+51.08**]
IDF1	48.59	**+5.74** [**+0.13**, **+13.11**]
IDS	1041	**−586** [**−951**, **−248**]
ByteTrack, YOLOv8n [5, 17]		
HOTA	29.05	**+1.77** [**+0.70**, **+2.94**]
MOTA	16.27	+1.02 [−1.06, +2.86]
IDF1	34.52	**+3.97** [**+1.88**, **+6.17**]
IDS	719	**−326** [**−691**, **−73**]

trackers by 4.15 to 5.89 HOTA points. Single-stage trackers gain through the rescue, and two-stage trackers when the detector is noisy.

Deployment. SCL needs no training, no GPU, and no access to the inside of the detector or the tracker. On a CPU it takes 2.95 ms per frame, small next to the 42 ms of the detector, and the tracker itself ran faster with it (1.43 against 2.04 ms per frame); the end-to-end time rose by only 4.5% (4-vCPU Intel Xeon at 2.10 GHz, no GPU, YOLOv8n and ByteTrack on three KITTI sequences, 2,279 timed frames). A new tracker is added through its four published contract values and an adapter, without changes to the rules.

7 Conclusion

This paper presented SCL, a training-free layer between an unchanged detector and an unchanged tracker. Nested Otsu splits of recent logits decide whether the scores are clean or noisy; on clean scores the layer keeps the tracker's scores and, where they fit, its thresholds, on noisy scores it chooses the candidates, rescores them, and sets the thresholds, and a continuation rescue gives single-stage trackers the step they lack. One frozen configuration restored two trackers from 0 to about 66 HOTA under halved scores, improved both single-stage trackers on every MOT17 sequence, raised MOTA by up to 24.70 points on held-out aerial data, reached higher HOTA than simple adaptive thresholds and ACT, kept well-matched trackers at their own accuracy

with no statistically detected HOTA loss, and took 2.95 ms per frame on a CPU. The score scale of the detector should therefore be treated as part of the tracking system. Future work will test the layer with more detectors, trackers, and datasets.

References

[1] Zhu, P., Wen, L., Du, D., Bian, X., Fan, H., Hu, Q., Ling, H.: Detection and Tracking Meet Drones Challenge. IEEE Transactions on Pattern Analysis and Machine Intelligence **44**(11), 7380–7399 (2022) https://doi.org/10.1109/tpami.2021.3119563

[2] Wu, X., Li, W., Hong, D., Tao, R., Du, Q.: Deep Learning for Unmanned Aerial Vehicle-Based Object Detection and Tracking: A survey. IEEE Geoscience and Remote Sensing Magazine **10**(1), 91–124 (2022) https://doi.org/10.1109/mgrs.2021.3115137

[3] Adžemović, M.: Deep learning-based multi-object tracking: a comprehensive survey from foundations to state-of-the-art. The Visual Computer **42**(9), 368 (2026) https://doi.org/10.1007/s00371-026-04607-7

[4] Bewley, A., Ge, Z., Ott, L., Ramos, F., Upcroft, B.: Simple online and realtime tracking. In: 2016 IEEE International Conference on Image Processing (ICIP), pp. 3464–3468 (2016). https://doi.org/10.1109/icip.2016.7533003

[5] Zhang, Y., Sun, P., Jiang, Y., Yu, D., Weng, F., Yuan, Z., Luo, P., Liu, W., Wang, X.: ByteTrack: Multi-object Tracking by Associating Every Detection Box. In: Computer Vision – ECCV 2022. Lecture Notes in Computer Science, pp. 1–21 (2022). https://doi.org/10.1007/978-3-031-20047-2_1

[6] Cao, J., Pang, J., Weng, X., Khirodkar, R., Kitani, K.: Observation-Centric SORT: Rethinking SORT for Robust Multi-Object Tracking. In: 2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 9686–9696 (2023). https://doi.org/10.1109/cvpr52729.2023.00934

[7] Küppers, F., Kronenberger, J., Shantia, A., Haselhoff, A.: Multivariate Confidence Calibration for Object Detection. In: 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), pp. 1322–1330 (2020). https://doi.org/10.1109/cvprw50498.2020.00171

[8] Luiten, J., Ošep, A., Dendorfer, P., Torr, P., Geiger, A., Leal-Taixé, L., Leibe, B.: HOTA: A Higher Order Metric for Evaluating Multi-object Tracking. International Journal of Computer Vision **129**(2), 548–578 (2021) https://doi.org/10.1007/s11263-020-01375-2

[9] Ma, L.V., Hussain, M.I., Park, J., Kim, J., Jeon, M.: Adaptive Confidence Threshold for ByteTrack in Multi-Object Tracking. In: 2023 12th International Conference on Control, Automation and Information Sciences (ICCAIS), pp. 370–374 (2023). https://doi.org/10.1109/iccais59597.2023.10382403

[10] Shim, K., Ko, K., Hwang, J., Kim, C.: AdapTrack: Adaptive Thresholding-Based Matching for Multi-Object Tracking. In: 2024 IEEE International Conference on Image Processing (ICIP), pp. 2222–2228 (2024). https://doi.org/10.1109/icip51287.2024.10648139

[11] Gwon, D.U., Lee, W.I.: Adaptive Threshold Control Scheme Based on Detection Confidence Distribution for Data Association in Multi-Object Tracking. Electronics **15**(17), 4047 (2026) https://doi.org/10.3390/electronics15174047

[12] Han, Y., Huang, G., Song, S., Yang, L., Wang, H., Wang, Y.: Dynamic Neural Networks: A Survey. IEEE Transactions on Pattern Analysis and Machine Intelligence **44**(11), 7436–7456 (2022) https://doi.org/10.1109/tpami.2021.3117837

[13] Jiang, J., Ananthanarayanan, G., Bodik, P., Sen, S., Stoica, I.: Chameleon: Scalable Adaptation of Video Analytics. In: Proceedings of the 2018 Conference of the ACM Special Interest Group on Data Communication, pp. 253–266 (2018). https://doi.org/10.1145/3230543.3230574

[14] Milan, A., Leal-Taixe, L., Reid, I., Roth, S., Schindler, K.: MOT16: A Benchmark for Multi-Object Tracking. arXiv preprint arXiv:1603.00831 (2016). https://doi.org/10.48550/arXiv.1603.00831

[15] Ge, Z., Liu, S., Wang, F., Li, Z., Sun, J.: YOLOX: Exceeding YOLO Series in 2021. arXiv preprint arXiv:2107.08430 (2021). https://doi.org/10.48550/arXiv.2107.08430

[16] Geiger, A., Lenz, P., Urtasun, R.: Are we ready for autonomous driving? The KITTI vision benchmark suite. In: 2012 IEEE Conference on Computer Vision and Pattern Recognition, pp. 3354–3361 (2012). https://doi.org/10.1109/cvpr.2012.6248074

[17] Jocher, G., Chaurasia, A., Qiu, J.: Ultralytics YOLOv8. https://github.com/ultralytics/ultralytics. Version 8.0.0 (2023)

[18] Zhao, Y., Lv, W., Xu, S., Wei, J., Wang, G., Dang, Q., Liu, Y., Chen, J.: DETRs Beat YOLOs on Real-time Object Detection. In: 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 16965–16974 (2024). https://doi.org/10.1109/cvpr52733.2024.01605

[19] Du, D., Qi, Y., Yu, H., Yang, Y., Duan, K., Li, G., Zhang, W., Huang, Q., Tian, Q.: The Unmanned Aerial Vehicle Benchmark: Object Detection and Tracking. In: Computer Vision – ECCV 2018. Lecture Notes in Computer Science, pp. 375–391. Springer, Cham (2018). https://doi.org/10.1007/978-3-030-01249-6_23

[20] Bernardin, K., Stiefelhagen, R.: Evaluating Multiple Object Tracking Performance: The CLEAR MOT Metrics. EURASIP Journal on Image and Video Processing **2008**, 1–10 (2008) https://doi.org/10.1155/2008/246309

[21] Ristani, E., Solera, F., Zou, R., Cucchiara, R., Tomasi, C.: Performance Measures and a Data Set for Multi-target, Multi-camera Tracking. In: Computer Vision – ECCV 2016 Workshops. Lecture Notes in Computer Science, pp. 17–35 (2016). https://doi.org/10.1007/978-3-319-48881-3_2

[22] Dendorfer, P., Ošep, A., Milan, A., Schindler, K., Cremers, D., Reid, I., Roth, S., Leal-Taixé, L.: MOTChallenge: A Benchmark for Single-Camera Multiple Target Tracking. International Journal of Computer Vision **129**(4), 845–881 (2021) https://doi.org/10.1007/s11263-020-01393-0

[23] Efron, B., Tibshirani, R.J.: An Introduction to the Bootstrap, (1993). https://doi.org/10.1007/978-1-4899-4541-9

[24] Li, Y., Liu, Y., Zhou, C., Xu, D., Tao, W.: A lightweight scheme of deep appearance extraction for robust online multi-object tracking. The Visual Computer **40**(3), 2049–2065 (2024) https://doi.org/10.1007/s00371-023-02901-2

[25] Bao, Y., Yu, Y., Qi, Y., Wang, Z.: Multiple object tracking with adaptive multi-features fusion and improved learnable graph matching. The Visual Computer **40**(4), 2279–2292 (2024) https://doi.org/10.1007/s00371-023-02916-9

[26] Zhang, Y., Liang, Y., Wang, J., Zhu, H., Wang, Z.: Enhanced multi-object tracking via embedded graph matching and differentiable Sinkhorn assignment: addressing challenges in occlusion and varying object appearances. The Visual Computer **41**(8), 6067–6085 (2025) https://doi.org/10.1007/s00371-024-03772-x

[27] Ding, Y., Wang, Y., Wang, L., Zhang, B., Li, J., Guo, L., Yang, Z., Liang, W.: Enhancing multi-object tracking efficiency through result-guided feature extraction and query filtering. The Visual Computer **41**(11), 8153–8167 (2025) https://doi.org/10.1007/s00371-025-03861-5

[28] Liang, Z., Xiao, G., Hu, J., Wang, J., Ding, C.: MotionTrack: rethinking the motion cue for multiple object tracking in USV videos. The Visual Computer **40**(4), 2761–2773 (2024) https://doi.org/10.1007/s00371-023-02983-y

[29] Sun, Z., Lv, Z., Xiong, M., Hu, W., Wei, G., Zhou, J., He, T., Meng, Q.: Online visual multi-object tracking via blind super-resolution and clustering learning. The Visual Computer **42**(4), 189 (2026) https://doi.org/10.1007/s00371-026-04382-5

[30] Aharon, N., Orfaig, R., Bobrovsky, B.-Z.: BoT-SORT: Robust Associations Multi-Pedestrian Tracking. arXiv preprint arXiv:2206.14651 (2022). https://doi.org/10.48550/arXiv.2206.14651

[31] Liu, Z., Wang, X., Wang, C., Liu, W., Bai, X.: SparseTrack: Multi-Object Tracking by Performing Scene Decomposition Based on Pseudo-Depth. IEEE Transactions on Circuits and Systems for Video Technology **35**(5), 4870–4882 (2025) https://doi.org/10.1109/tcsvt.2024.3524670

[32] Stanojevic, V.D., Todorovic, B.T.: BoostTrack: boosting the similarity measure and detection confidence for improved multiple object tracking. Machine Vision and Applications **35**(3) (2024) https://doi.org/10.1007/s00138-024-01531-5

[33] Wang, Y., Zhang, D., Li, R., Zheng, Z., Li, M.: PD-SORT: Occlusion-Robust Multi-Object Tracking Using Pseudo-Depth Cues. IEEE Transactions on Consumer Electronics **71**(1), 165–177 (2025) https://doi.org/10.1109/tce.2025.3541839

[34] Yang, M., Han, G., Yan, B., Zhang, W., Qi, J., Lu, H., Wang, D.: Hybrid-SORT: Weak Cues Matter for Online Multi-Object Tracking. Proceedings of the AAAI Conference on Artificial Intelligence **38**(7), 6504–6512 (2024) https://doi.org/10.1609/aaai.v38i7.28471

[35] Jung, H., Kang, S., Kim, T., Kim, H.: ConfTrack: Kalman Filter-based Multi-Person Tracking by Utilizing Confidence Score of Detection Box. In: 2024 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), pp. 6569–6578 (2024). https://doi.org/10.1109/wacv57701.2024.00645

[36] Solano-Carrillo, E., Sattler, F., Alex, A., Klein, A., Costa, B.P., Rodriguez, A.B., Stoppe, J.: UTrack: Multi-object Tracking with Uncertain Detections. In: Computer Vision – ECCV 2024 Workshops. Lecture Notes in Computer Science, pp. 219–236. Springer, Cham (2025). https://doi.org/10.1007/978-3-031-91585-7_14

[37] Kim, S., Lee, S., Cho, M.: PlugTrack: Multi-Perceptive Motion Analysis for Adaptive Fusion in Multi-Object Tracking. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 40, pp. 5709–5716 (2026). https://doi.org/10.1609/aaai.v40i7.37491

[38] Wang, C.-Y., Yeh, I.-H., Liao, H.-Y.M.: YOLOv9: Learning What You Want to Learn Using Programmable Gradient Information. In: Computer Vision – ECCV 2024. Lecture Notes in Computer Science, pp. 1–21 (2024). https://doi.org/10.1007/978-3-031-72751-1_1

[39] Carion, N., Massa, F., Synnaeve, G., Usunier, N., Kirillov, A., Zagoruyko, S.: End-to-End Object Detection with Transformers. In: Computer Vision – ECCV 2020. Lecture Notes in Computer Science, pp. 213–229 (2020). https://doi.org/10.1007/978-3-030-58452-8_13

[40] Otsu, N.: A Threshold Selection Method from Gray-Level Histograms. IEEE Transactions on Systems, Man, and Cybernetics **9**(1), 62–66 (1979) https://doi.org/10.1109/tsmc.1979.4310076

[41] Reddy, B.S., Chatterji, B.N.: An FFT-based technique for translation, rotation, and scale-invariant image registration. IEEE Transactions on Image Processing **5**(8), 1266–1271 (1996) https://doi.org/10.1109/83.506761